P62 | N17

set of vectors which contain zero vector, is dependent.

N 29

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

p 79 N1

$T: R^2 \to R^4$

$$T(e_1) = A \cdot e_1 = \begin{bmatrix} 3 \\ 1 \\ 3 \\ 1 \end{bmatrix}$$

$$T(e_2) = A \cdot e_2 = \begin{bmatrix} -5 \\ 2 \\ 0 \\ 0 \end{bmatrix}$$

matrix:
$$\begin{bmatrix} 3 & -5 \\ 1 & 2 \\ 3 & 0 \\ 1 & 0 \end{bmatrix}$$

N 3

$$T=\begin{bmatrix}\cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi\end{bmatrix}=\begin{bmatrix}0 & 1 \\ -1 & 0\end{bmatrix} \quad \text{or } \varphi=\frac{3\pi}{2}$$

N 7

$x'=x=-\frac{\sqrt{2}}{2}$ $\quad x=\cos\left(-\frac{3\pi}{4}\right)=-\frac{\sqrt{2}}{2}$

$y'=-y=\frac{\sqrt{2}}{2}$ $\quad y=\sin\left(-\frac{3\pi}{4}\right)=-\frac{\sqrt{2}}{2}$

$$T(e_1)=\begin{bmatrix}-\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}}\end{bmatrix}$$

$x=\cos\left(-\frac{\pi}{2}\right)=\frac{1}{\sqrt{2}}$ $\quad x'=x=\frac{1}{\sqrt{2}}$

$y=\sin\left(-\frac{\pi}{2}\right)=-\frac{1}{\sqrt{2}}$ $\quad y'=-y=\frac{1}{\sqrt{2}}$

$$T(e_2)=\begin{bmatrix}\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}}\end{bmatrix} \quad \text{Matrix is} \begin{bmatrix}-\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{bmatrix}$$

N 15

$$\begin{bmatrix} ? & ? & ? \\ ? & ? & ? \\ ? & ? & ? \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3x_1 - 2x_3 \\ 4x_1 \\ x_1 - x_2 + x_3 \end{bmatrix}$$

$$?x_1 + ?x_2 + ?x_3 = 3x_1 - 2x_3$$

$$[3 \; 0 \; -2]$$

$$?x_1 + ?x_2 + ?x_3 = 4x_1$$

$$[4 \; 0 \; 0]$$

$$?x_1 + ?x_2 + ?x_3 = x_1 - x_2 + x_3$$

$$[1 \; -1 \; 1]$$

ans:

$$\begin{bmatrix} 3 & 0 & -2 \\ 4 & 0 & 0 \\ 1 & -1 & 1 \end{bmatrix}$$

N 17

$$T(x_1, x_2, x_3, x_4) = \begin{bmatrix} 0 \\ x_1 + x_2 \\ x_2 + x_3 \\ x_3 + x_4 \end{bmatrix}$$

$$\begin{bmatrix} ? & ? & ? & ? \\ ? & ? & ? & ? \\ ? & ? & ? & ? \\ ? & ? & ? & ? \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ x_1 + x_2 \\ x_2 + x_3 \\ x_3 + x_4 \end{bmatrix}$$

Matrix is

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$